

1 Acquisition

views per plant
ordering (video or
discrete views)
pose, depth,
baseline, sides

2 Observation

detections or
masks in each
image; one
fruit may yield
several boxes

3 Association

decide which
observations belong
to the same fruit
(six mechanisms
below)

4 Attribute assignment

class label, size,
mass assigned to
the unique
instance, not to
each box

5 Inventory

one record per
physical fruit,
aggregated by
class, tree, row,
or block



No association

Assumes:
each fruit is seen
once, or duplicates
are ignored

Statistical correction

Assumes:
duplication is stable
enough to calibrate
against hand counts

Appearance matching

Assumes:
fruit look different
enough to be matched
by appearance

Temporal tracking

Assumes:
consecutive frames
overlap; motion is
smooth and bounded

Geometric or 3D association

Assumes:
pose, scale, and
depth or reconstruction
are reliable

Learned association

Assumes:
linked identities
exist to train and
test the matcher